

Program: Diploma in Microelectronics	
Course Code: 6491D	Course Title: VLSI for Wireless Communication
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To understand the concepts of basic wireless communication concepts.
- To study the parameters in receiver and low noise amplifier design.
- To study the various types of mixers designed for wireless communication.
- To study and design PLL and VCO.
- To understand the concepts of transmitters and power amplifiers in wireless communication.

Course Prerequisites:

Nil

Course Outcomes:

After the completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Explain basic wireless communication concepts.	12	Understanding
CO2	To understand the parameters in receiver and design a low noise amplifier	14	Understanding
CO3	Explain various types of mixers designed for wireless communication.	12	Understanding
CO4	Design of PLL and VCO, also understand the concepts of transmitters and utilize the power amplifiers in wireless communication.	20	Applying
	Series Tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2	2				
CO3	3	2					
CO4	3	3	2				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain basic wireless communication concepts.		
M1.01	Explain Basic wireless communication systems	3	Understanding
M1.02	Describe various types of modulation schemes used	3	Understanding
M1.03	Explain the concept of pathloss and fading	6	Understanding
Contents: COMMUNICATION CONCEPTS : Introduction – Overview of Wireless systems – Standards – Access Methods – Modulation schemes – Classical channel – Wireless channel description – Path loss – Multipath fading.			
CO2	To understand the parameters in receiver and design a low noise amplifier		
M2.01	Explain receiver architecture	2	Understanding
M2.02	Explain Noise figure and design parameters	5	Understanding
M2.03	Describe filters and its design	3	Understanding
M2.04	Design different types of LNA	4	Applying
	Series Test 1	1	

Contents : RECEIVER ARCHITECTURE & LOW NOISE AMPLIFIERS Receiver front end – Filter design – Non-idealities – Design parameters – Noise figure & Input intercept point. LNA Introduction – Wideband LNA design – Narrow band LNA design: Impedance matching & Core amplifier.			
CO3	Explain various types of mixers designed for wireless communication.		
M3.01	Describe Various types of Mixers	4	Understanding
M3.02	Explain the working principle of unbalanced mixer	4	Understanding
M3.03	Explain the working principle of single ended sampling mixer	4	Understanding
Contents: MIXERS: Balancing Mixer - Qualitative Description of the Gilbert Mixer - Conversion Gain – Distortion – Noise - A Complete Active Mixer. Switching Mixer – Distortion, Conversion Gain & Noise in Unbalanced Switching Mixer - A Practical Unbalanced Switching Mixer. Sampling Mixer - Conversion Gain, Distortion, Intrinsic & Extrinsic Noise in Single Ended Sampling Mixer.			
CO4	Design of DPLL and VCO, also understand the concepts of transmitters and utilize the power amplifiers in wireless communication.		
M4.01	Explain Frequency synthesizers	4	Understanding
M4.02	Explain DPLL	4	Understanding
M4.03	Explain Design approaches in DPLL	5	Understanding
M4.04	Describe power amplifiers	7	Understanding
	Series Test 2	1	
Contents : FREQUENCY SYNTHESIZERS & DPLL – Phase detector – Dividers – Voltage Controlled Oscillators – LC oscillators – Ring Oscillators – Phase noise – Loop filters & design approaches – A complete synthesizer design example (DECT) – Frequency synthesizer with fractional divider. TRANSMITTER ARCHITECTURES & POWER AMPLIFIERS: Transmitter back end design – Quadrature LO generator – Power amplifier design.			

Text/Reference:

T/R	Book Title/Author
T1	Bosco H Leung “VLSI for Wireless Communication”, Pearson Education, 2002
T2	B.Razavi ,”RF Microelectronics” , Prentice-Hall ,1998
R1	Behzad Razavi, “Design of Analog CMOS Integrated Circuits” McGraw-Hill, 1999.
R2	Emad N Farag and Mohamed I Elmasry, “Mixed Signal VLSI wireless design – Circuits & Systems”, Kluwer Academic Publishers, 2000.
R3	J. Crols and M. Steyaert, “CMOS Wireless Transceiver Design,” Boston, Kluwer Academic Pub., 1997.
R4	Thomas H.Lee, “The Design of CMOS Radio – Frequency Integrated Circuits”, Cambridge University Press ,2003.